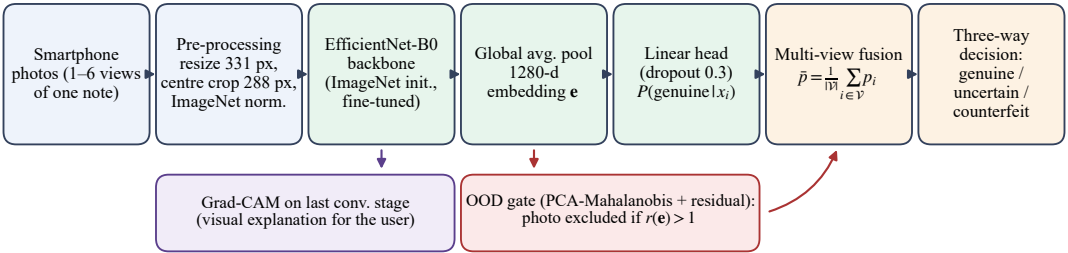
Smartphone photos (1–6 views of one note)
Pre-processing resize 331 px, centre crop 288 px, ImageNet norm.
EfficientNet-B0 backbone (ImageNet init., fine-tuned)
Global avg. pool 1280-d embedding e
Linear head (dropout 0.3) P(genuine|x_i)
Multi-view fusion p̄ = 1/|V| Σ_{i∈V} p_i
Three-way decision: genuine / uncertain / counterfeit
Grad-CAM on last conv. stage (visual explanation for the user)
OOD gate (PCA-Mahalanobis + residual): photo excluded if r(e) > 1